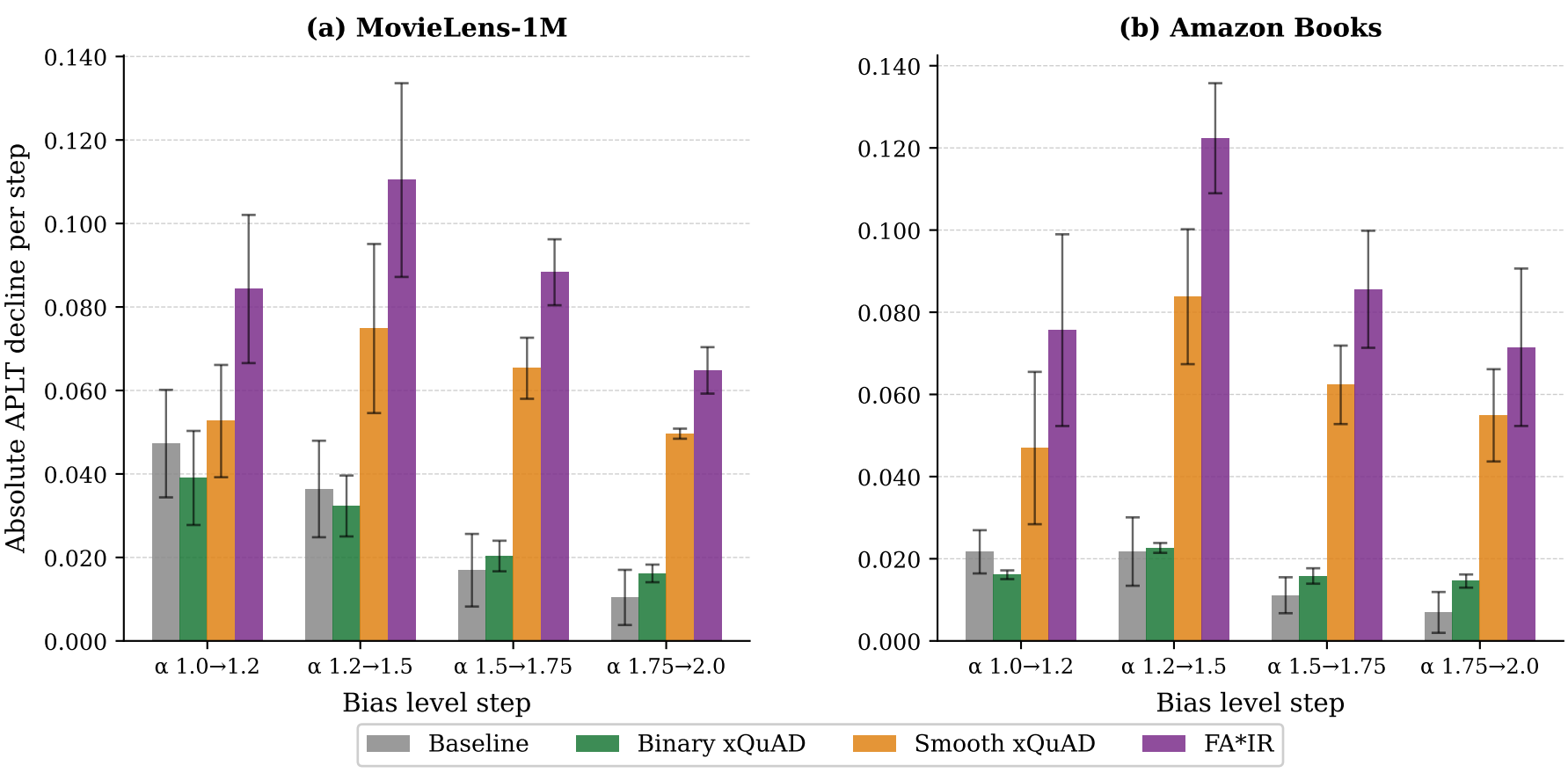

Error bars show ±1 std across BPR, NeuMF and LightGCN. Larger bars indicate faster structural ceiling collapse at that step.